

Lesson 1.2: How We Get Oxygen – The Lungs' Tiny Air Sacs

What You'll Learn

By the end of this lesson, you'll understand how air gets into the lungs and how oxygen enters the blood.

The Content

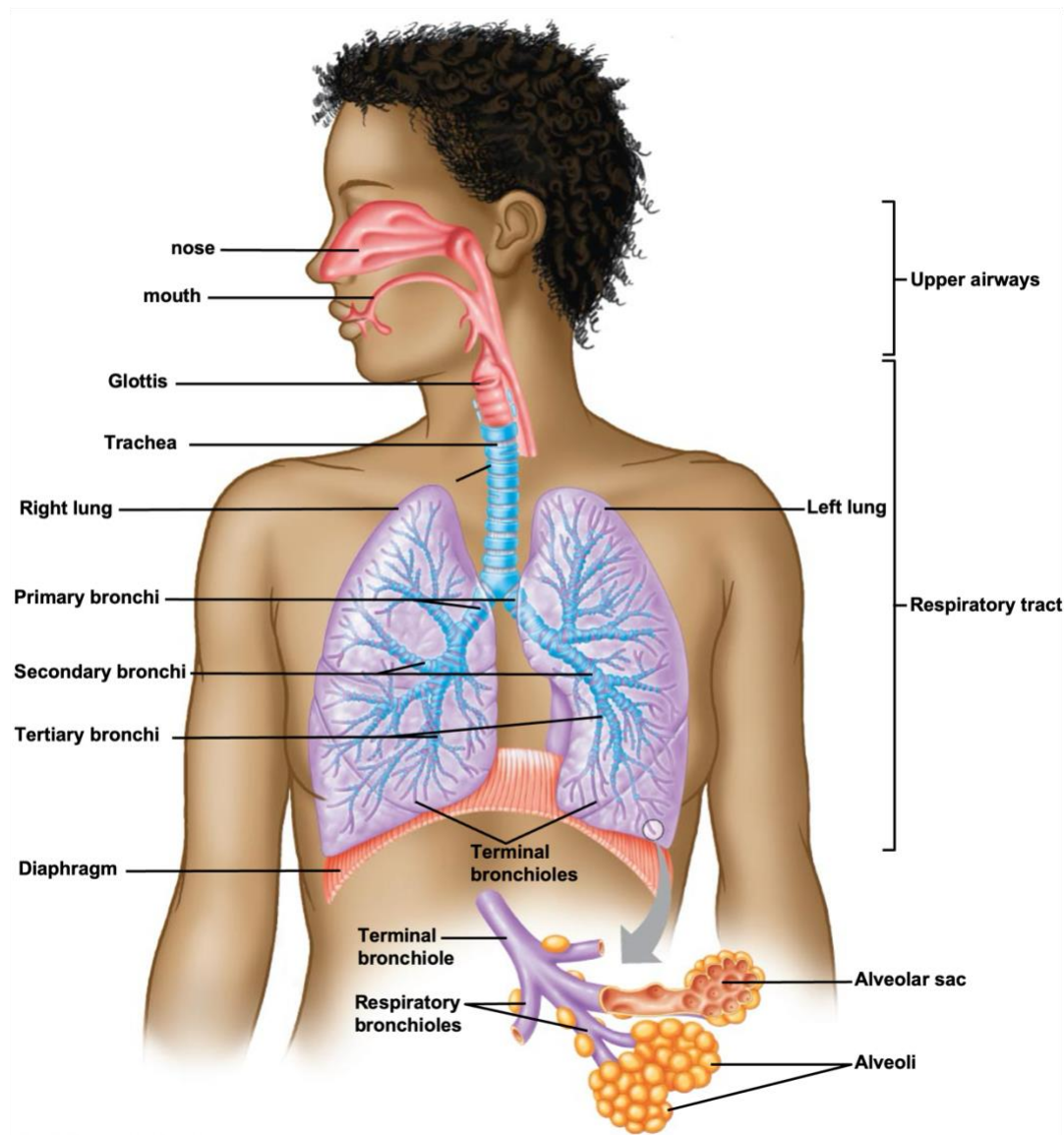
The journey of air into the body

Breathing starts with a muscle called the diaphragm (say: DY-ah-fram). The diaphragm sits below your lungs, separating your chest from your belly. It's shaped like a dome, sort of like an upside-down bowl.

When you need to breathe in, your brain sends a signal to the diaphragm. The diaphragm contracts (tightens up) and flattens out, moving downward. This pulls on the lungs and makes them expand, creating more space inside your chest.

When there's more space in the chest, the pressure inside drops below the pressure outside your body. Air naturally flows from areas of higher pressure to lower pressure, so air rushes in through your nose or mouth to fill that space. It's similar to how water flows downhill – it naturally moves to fill empty spaces.

When you breathe out, the diaphragm relaxes and moves back up into its dome shape. This pushes air out of the lungs.



When we breathe in, air follows this path:

1. Air enters through the nose or mouth
2. It travels down the throat, through the trachea (say: TRAY-kee-ah), also called the windpipe
3. The trachea splits into two large tubes called bronchi (say: BRON-kye) – one for each lung
4. The bronchi branch into smaller and smaller tubes called bronchioles (say: BRON-kee-olz)

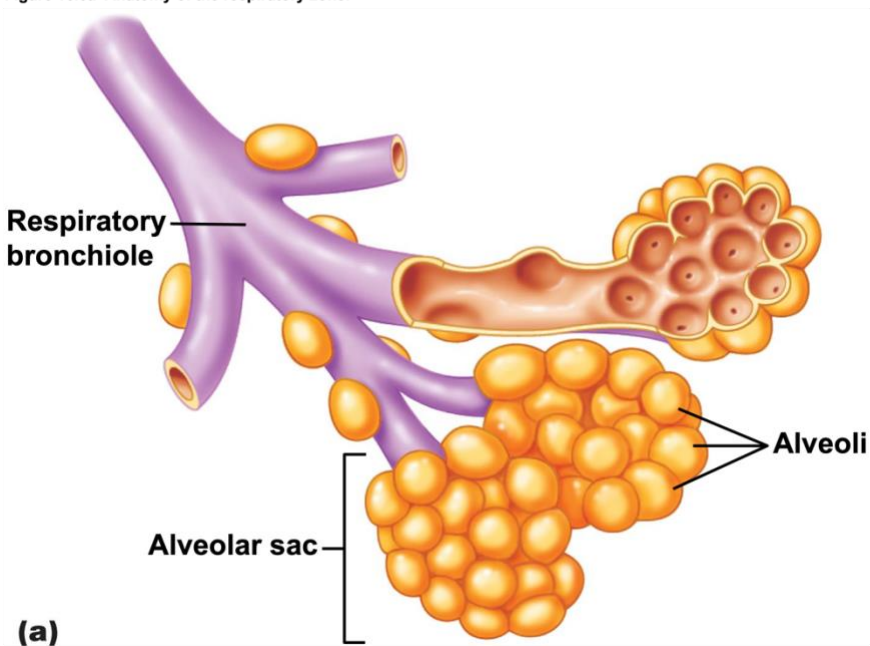
5. The smallest tubes, called respiratory bronchioles, lead to millions of tiny air sacs called alveoli (say: al-VEE-oh-lie)

This whole process of moving air in and out of the lungs is sometimes called ventilation or external respiration.

What are alveoli?

Alveoli are like tiny balloons inside your lungs. Your lungs have millions of them – so many that if you laid them all flat, they would cover a tennis court!

Figure 16.5a Anatomy of the respiratory zone.



The walls of the alveoli are made of special cells:

- **Type I alveolar cells** – These are very thin, flat cells that cover most of the alveoli surface. They are so thin that oxygen and carbon dioxide can easily pass through them.
- **Type II alveolar cells** – These are the "factory" cells that make surfactant. We'll learn more about them in Lesson 1.4.

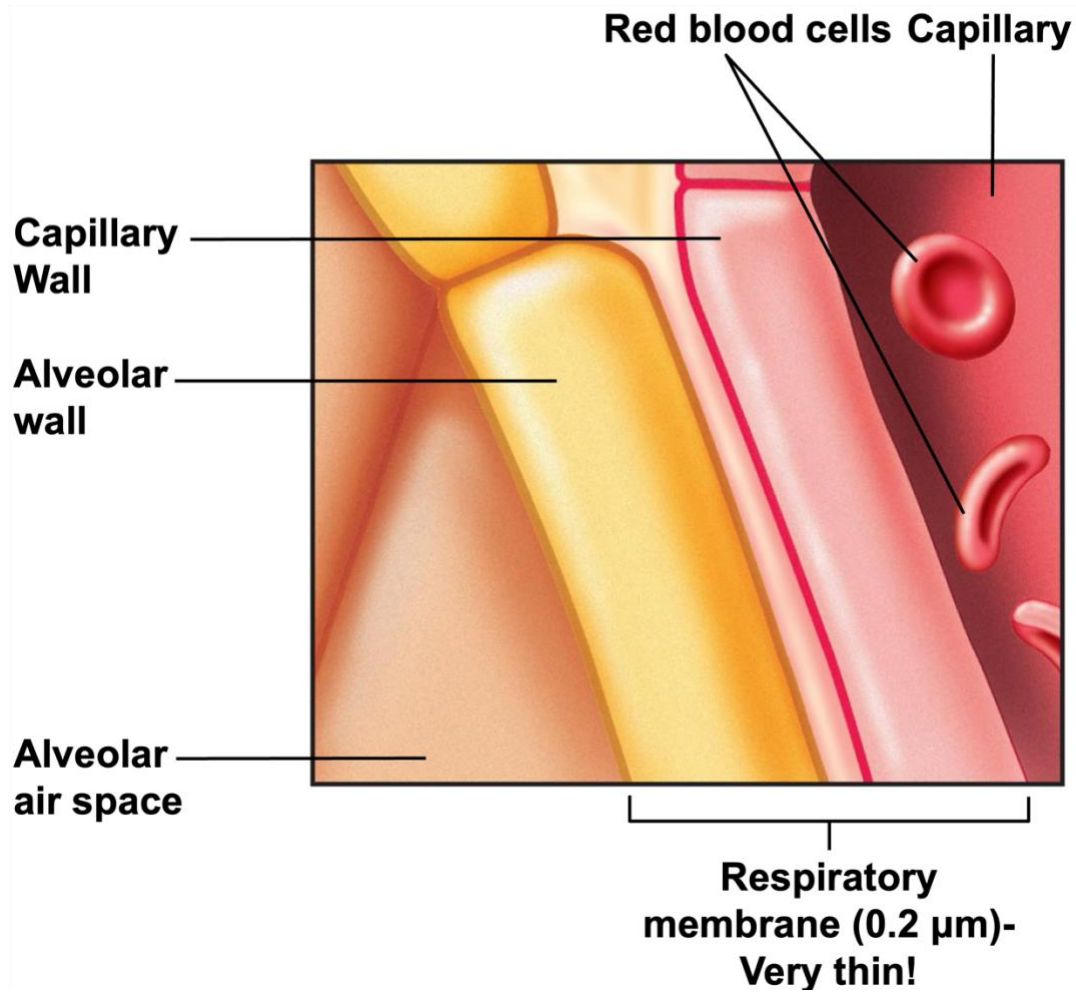
- **Macrophages** (say: MACK-row-fay-jez) – These are like cleanup crews or security guards. They patrol the alveoli, eating up germs, dust, and old surfactant to keep the lungs clean and healthy.

These tiny air sacs are where the magic happens. They are where oxygen gets into your blood.

How does oxygen get into the blood?

The wall of each alveolus is very, very thin – thinner than a piece of tissue paper. This thin wall is called the respiratory membrane. It sits right next to tiny blood vessels called

capillaries (say: CAP-ih-lair-eez).



When air fills the alveolus, oxygen passes through the respiratory membrane and enters the blood in the capillaries. At the same time, carbon dioxide (a waste gas from your body) leaves the blood and enters the alveolus to be breathed out.

This trade – oxygen in, carbon dioxide out – is called "gas exchange." It happens with every breath, over 30,000 times a day!

Key Takeaway: The lungs contain millions of tiny air sacs called alveoli. These are where oxygen enters the blood and carbon dioxide leaves the body.